Sign Errors in Parabola Transformations
Answer Key

Algebra 1 Quadratic Functions

Quick Answers

1. 2	3. -3, 5	5. 0	7. -3, 7
2. -3	4. 7	6. -5	8. —

Common wrong answers

1. *If they got 102: read x - 5 as a vertex at x = -5 instead of x = 5*
2. *If they got 61: wrote (x - 4) squared, minus 3, for a shift LEFT 4, then evaluated at x = -4*
4. *If they got -137: substituted x = -6, reading the sign inside the parentheses literally*
5. *If they got 24: dropped the negative on the leading coefficient, so the reflection was lost*
6. *If they got 191: read x + 7 as a shift RIGHT 7 and evaluated the minimum at x = 7*

Grading note. On this sheet the sign reasoning is the skill, so a correct number reached by the wrong substitution should not earn full credit. The "common wrong answers" list above tells you which misreading produced each wrong value.

1. Minimum value of $f(x) = (x-5)^2 + 2$.

The vertex x is whatever makes the squared factor zero: $x - 5 = 0$ gives $x = 5$, so the vertex is at $x = 5$, *not* $x = -5$. (The graph moved right, in the opposite direction from the printed sign.)

$$f(5) = (5-5)^2 + 2 = 0 + 2$$

Because $(x-5)^2 \geq 0$ for every x, adding 2 can never produce anything smaller, so this really is the minimum. $\boxed{2}$

If they got 102: they substituted $x = -5$, reading the printed sign literally.

2. $y = x^2$ shifted 4 left and 3 down; evaluate at $x = -4$.

A shift left is written with a *plus* inside the parentheses, because the graph moves to where $x + 4 = 0$, i.e. $x = -4$:

$$y = (x+4)^2 - 3$$

$$y(-4) = (-4+4)^2 - 3 = 0 - 3$$

The value at $x = -4$ is the vertex height, which is the whole point of the check. $\boxed{-3}$

If they got 61: they wrote $(x-4)^2 - 3$ for "left 4" and then evaluated it at $x = -4$, so the two sign choices compounded.

3. x-intercepts of $p(x) = (x-1)^2 - 16$.

Set $p(x) = 0$ and undo the transformations in reverse order:

$$\begin{aligned} (x-1)^2 - 16 &= 0 \\ (x-1)^2 &= 16 \\ x - 1 &= \pm 4 \qquad \text{(both roots — dropping } -4 \text{ loses an intercept)} \\ x = 1 + 4 = 5 &\qquad \text{or} \qquad x = 1 - 4 = -3 \end{aligned}$$

Sign check: the axis of symmetry is $x = 1$, and the two intercepts sit 4 units on either side of it ✓

$$\boxed{x = -3 \quad \text{and} \quad x = 5}$$

4. Find and fix: maximum of $y = -(x-6)^2 + 7$.

The error: Jamal read $(x-6)$ as a vertex at $x = -6$. The vertex sits where the squared factor is zero, so $x - 6 = 0$ gives $x = 6$. His arithmetic was fine; his h was not.
Correct substitution:

$$y(6) = -(6-6)^2 + 7 = -0 + 7$$

Since $a = -1 < 0$ the parabola opens downward, so this vertex value is a *maximum* — 7 is the largest output the function ever reaches. $\boxed{7}$
If they got -137*:* they repeated Jamal's substitution $x = -6$ instead of fixing it.

5. Reflect, stretch by 3, left 2, up 12; evaluate at $x = 0$.

Build the equation one transformation at a time. Reflection and vertical stretch both live in a: $a = -3$. Left 2 puts $+2$ inside the parentheses; up 12 adds 12 outside.

$$q(x) = -3(x+2)^2 + 12$$

$$q(0) = -3(0+2)^2 + 12 = -3(4) + 12 = -12 + 12$$

So $x = 0$ is one of the x-intercepts of the transformed parabola. $\boxed{0}$
If they got 24: they kept $a = +3$ and lost the reflection. Note the vertex is at $x = -2$ with value 12, so a positive a would make 12 a minimum instead of a maximum — the shape of the answer changes, not just the number.

6. Find and fix: minimum of $g(x) = (x+7)^2 - 5$.

The error: Mia called $(x+7)$ a shift 7 units *right*. It is a shift 7 units *left*, because $x + 7 = 0$ at $x = -7$. Her "down 5" was correct.
Because she put the vertex at $x = 7$, she evaluated at the wrong place. Correct substitution:

$$g(-7) = (-7+7)^2 - 5 = 0 - 5$$

$\boxed{-5}$
If they got 191: that is $g(7)$ — Mia's value. It is a real point on the parabola, just not the lowest one, which is exactly why a wrong h can go unnoticed.

7. Reflect, right 2, up 25; find the x-intercepts.

Reflection makes $a = -1$; right 2 gives $(x - 2)$; up 25 adds 25:

$$y = -(x - 2)^2 + 25$$

Set it to zero:

$$\begin{aligned} -(x-2)^2 + 25 &= 0 \\ (x-2)^2 &= 25 \\ x - 2 &= \pm 5 \\ x &= 7 \qquad \text{or} \qquad x = -3 \end{aligned}$$

$$\boxed{x = -3 \quad \text{and} \quad x = 7}$$

Why the reflection does not move the axis: reflecting over the x-axis replaces each output y with $-y$ and leaves every input untouched. The axis of symmetry is a statement about inputs, so it stays at $x = 2$ — and indeed -3 and 7 are each 5 units from 2.

8. Error analysis — Ravi's study card. (open response — read the explanations)

R1 is wrong. The shift is not in the direction of the printed sign. The vertex lands where the squared factor is zero, so $(x + 4)$ puts the vertex at $x = -4$: the graph moves *left* 4. *Corrected:* "In $y = a(x - h)^2 + k$ the graph shifts to $x = h$, the value that makes $x - h$ zero. So $(x + 4)$ means $h = -4$: left 4."

R2 is correct. k is added to every output after the squaring, so a positive k raises the whole graph and a negative k lowers it. Here the printed sign and the direction agree, because nothing is subtracted *from* y.

R3 is wrong. a multiplies the outputs; it does nothing to the inputs. A negative a flips the parabola upside down about its own vertex, but the vertex — and therefore both shifts — stays exactly where h and k put it. *Corrected:* "When $a < 0$ the parabola opens downward and its vertex becomes a maximum instead of a minimum; the shifts are unchanged."

Full credit requires naming R1 and R3 as wrong, an explanation that appeals to "what makes the parentheses zero" (R1) and to "a scales outputs, not inputs" (R3), and a corrected wording for each. Naming the wrong rules with no reason is partial credit. $\boxed{\text{R1 wrong, R2 correct, R3 wrong}}$